MILITARY SPECIFICATION

TOURNIQUET, NONPNEUMATIC, CAMOUFLAGED, 1½ BY 42 INCHES

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 This specification covers a web tourniquet for use in the control of hemorrhages from an extremity by the application of circumferential mechanical pressure

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

V-T-276 — Thread Cotton
CCC-T-191 — Textile Test Methods
PPP-B-636 — Box Fiberboard
PPP-T-45 — Tape, Gummed, Paper, Reinforced
PPP-T-76 — Tape Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing)

MILITARY

MIL-W-530 — Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-T-3530 — Treatments, Mildew Resistant and Water Repellent for Thread and Twine

STANDARDS

FEDERAL

Fed Std No 751 — Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 — Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply

COLOR ASSOCIATION OF THE UNITED STATES, INC

United States Army Standard Color Card for the Official Standardized Shades of Sewing Threads

FSC 6515

571076—61—(X-149)

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(Application for copies should be addressed to the Color Association of the United States, Inc, 200 Madison Avenue, New York 16, N Y

3. REQUIREMENTS

3.1 Materials Except as specified herein, materials shall conform to the applicable portions of the specifications and standards listed in section 2, and shall be capable of meeting the requirements shown therein When materials to be used are not specifically designated, those used shall be of the highest commercial quality, and entirely suitable for the purpose intended

3.1.1 *Strap* Strap shall be fabricated of cotton webbing composed of two ground warps (one face, one back), one binder warp, and stuffer warp, as follows

Face warp — Face warp shall be plain weave with picks that show in the face

Back warp — Back warp shall be plain weave with picks that show in the back

Binder warp — Binder warp shall be plain weave

There shall be 4 face ends and 2 back ends between each binder, 16 ground threads on one edge and 17 on the other, and 2 stuffer threads between each binder and 2 on each edge Physical characteristics of the webbing shall be as follows

Width	1½ inch (±$\frac{1}{32}$ inch)
Minimum weight per linear yard	0 52 ounce
Minimum warp threads per inch	191
Face and edge per inch	101
Back per inch	34
Binder per inch	18
Stuffer per inch	38
Filling per inch	48
Minimum breaking strength	300 pounds

Approximate yarn sizes.

Face and edge	60/2
Back	20/2
Binder	30/2
Stuffer	12/2
Filling	20/2

Strap shall conform to Specification MIL-W-530 for class 4 finish requirements only, except that the maximum amount of copper—8—quinolinolate shall be 1 28 percent (see 4 4 1) The strap shall be vat dyed olive drab No 7

3.1.2 *Buckle rest* Buckle rest shall be fabricated of cotton twill having a 2 by 2 herringbone twill weave with three reversals of twill equidistant across the web The twill shall have the following physical characteristics

Minimum weight per linear yard	1 25 ounces
Minimum warp threads	168
Filling per inch	19
Minimum breaking strength per inch width	210 pounds

Approximate yarn sizes

Warp	10/3
Filling	20/4

Buckle rest shall conform to Specification MIL-W-530 for class 4 finish requirements only, except that the maximum amount of copper—8—quinolinolate shall be 1 28 percent (see 4 4 1) The buckle rest shall be vat dyed olive drab No 7

3.1.3 *Buckle* The buckle shall be a readily released spring-tension, clamp type buckle for maintenance of tourniquet pressure Buckle shall be fabricated of brass bronze, or steel and shall be of the types as shown in figure 1 and of the dimensions as to properly accommodate the 1½-inch-wide webbing (see 6 1) All buckles shall have a black oxidized finish

3.1.4 *Thread* Except as specified herein, the thread for stitching shall be in accordance with the requirements of Specification V-T-276 for type IIIA, heavy thread 3-ply ticket No 10/3, as shown in table V for bobbin thread, and type IIIB, heavy thread, 3-ply ticket No 10/3 as shown in table VI for needle thread The thread shall be vat dyed olive drab Shade S-1 conforming to U S Army Standard Color Card All references to "approved shade standard", "approved standard shade sample", and "approved standard shade", made in Specification V-T-276 shall not apply Thread shall be treated for mildew-resistance in accordance with Specification MIL-T-3530, using type I, class 1 mildew-inhibiting agent (see 4 4 2) The finished thread shall show

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fair colorfastness to laundering, chlorine bleaching, and weather Paragraph titled "Certification" in Specification V-T-276 shall not apply (see 4 1 2)

3.1.5. *Label* The label shall be fabricated of a suitable cotton fabric, shall be vat dyed olive drab No 7, and shall meet the finish requirements specified herein for the strap and buckle rest

3.2 Construction. Web tourniquet shall be constructed as shown in figure 1 The webbing shall be 42 inches (plus ½ inch minus zero) long and 1½ inches (plus or minus $\frac{1}{32}$ inch) wide The buckle rest shall consist of two layers of herringbone twill sewn together (with the ends of each layer turned under), so that the finished pad is of the dimensions as shown in the figure The layers of twill shall be sewn together by stitching around the entire outer edge and by stitching two diagonal lines through the pad One end of the web strap shall be double looped around the buckle rod, and the strap end containing the buckle shall be sewn to the buckle rest by at least four rows of stitching in a manner as shown in the figure All stitching shall conform to Federal Standard 751 and consist of stitch type 301 stitches There shall be eight to ten stitches per inch The free end of the web strap shall be reduced in dimension by cutting off one corner of the end with a diagonal cut 1½ inches long, so that the resultant width of the end of the strap shall be ½ inch The diagonally cut free end of the strap shall be finished by being dipped in a vinyl resin for the entire length of the cut and shall be capable of passing the tests specified in 4 4 1 3

3.3 Strap and buckle.

3.3.1 *Strap slippage* There shall be no evidence of slippage when the strap is inserted in the buckle and subjected to a 130-pound pull The teeth of the buckle shall be such as to give an even hold over the entire width of the strap Strap shall be tested as specified in 4 4 4

3.3.2 *Initial force required to open buckle* The initial force required to open the buckle shall be 3±½ pounds Buckle shall be tested as specified in 4 4 3

3.4 Instruction label and identification marking. A label containing instructions for the use of the tourniquet shall be sewn to the outside surface of the strap (surface away from patient) 3¾ inches away from the buckle The label shall be securely attached to the strap by stitching around its four sides The following legend shall be permanently and legibly printed with red ink on the label

> "CAUTION—Do not apply tourniquet unless blood is gushing Tighten only enough to stop arterial bleeding Tourniquet should not be loosened by anyone except a medical officer or medically trained personnel equipped to control the bleeding by other means Repeated loosening is dangerous"

This label shall also be printed with the manufacturer's name or registered trade-mark which shall appear below the legend specified above

3.5 Workmanship. Workmanship shall be first class throughout Tourniquets shall be free from defects which detract from their appearance or may impair their serviceability

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, etc, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained,

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may be furnished in lieu of actual performance of such testing by the contractor.

4.1.3 Government material shall not be sampled or tested unless specifically requested by the Government (see 6.2)

4.2 Sampling.

4.2.1 *For examination* Sampling for examination shall be conducted in accordance with Standard MIL-STD-105

4.2.2 *For tests* Sampling for tests shall be conducted in accordance with Standard MIL-STD-105

	Inspection level	AQL (percent defective)	Minimum sample size
Component material testing	L-2	*	3
End item testing	L-2	*	3

* Acceptance number shall be zero (0) for all sample sizes

4.2.3 *Component materials* Sufficient samples of the component materials shall be selected from each lot of material used in the manufacture of the tourniquets to perform tests as required by the applicable subsidiary specifications or by tests specified herein When sample size for component materials covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated in 4.2.2 The unit of product for testing shall be the quantity of material necessary to perform all required tests one time each, taking into consideration the fact that, where possible the same material may be used to perform two or more tests Lot sizes are expressed in multiples of the following

Webbing	Yards or fractions of a yard
Thread	Cone or spool
Buckle	Each

4.3 Examination. The tourniquets shall be examined to determine compliance with all requirements contained in this specification

4.4 Tests. Tests shall be conducted, as necessary to determine compliance with specification requirements

4.4.1 *Webbing, twill and label fabric* The webbing, twill, and label fabric shall be tested for compliance with the requirements of this specification, in accordance with the test procedures of Specification MIL-W-530, as applicable, except that the performance of the tests for mildew-inhibitor content and mildew-resistance shall be conducted under the following conditions:

(a) If the material conforms to the requirements for mildew-inhibitor content, the mildew-resistance tests shall not be performed, and the material shall not be considered as complying with mildew-resistance requirements

(b) If the mildew-inhibitor content exceeds the maximum requirement, the material shall be rejected

(c) If the mildew-inhibitor content is found to be below the minimum requirement, the material may be accepted with respect to mildew-resistance provided it passes the mildew-resistance test

4.4.1.2 *Mildew-resistance* The finished webbing, twill and label fabric shall be tested by method 4762 of Specification CCC-T-191 except that the leaching process shall be omitted

4.4.1.3 *Webbing vinyl-dip finish* The vinyl-resin of the diagonally-cut free end of the straps shall not become tacky after being conditioned for 4 hours at 180° Fahrenheit (F.), shall not crack or peel off when bent 180° over a 4-inch mandrel at minus 60° F. (after being at this temperature for 1 hour), and shall not swell up or become gelatinous after being conditioned for 4 hours in an atmosphere of 100-percent relative humidity at 100° F. In addition after being subjected to accelerated aging in accordance with method 5852 of Specification CCC-T-191, the vinyl-resin dipped end of the straps shall be capable of passing the tests described above

4.4.2 *Thread* The untreated thread shall be tested in accordance with the test procedures of Specifications V-T-276 (see 3.1.4) The treated thread shall be tested as specified in the test section of Specification MIL-T-3530

4.4.2.1 *Colorfastness to laundering* The finished thread shall be tested for colorfastness to laundering by method 4610 of Specification CCC-T-191

4.4.3 *Initial force required to open buckle* The initial force required to open the buckle shall be tested with a Chatillon Dynamometer No

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80D The end of the rod shall be pressed against the end of the thumbpiece and pressure applied The initial force required to open the buckle shall be as specified in 3 3 2

4.4.4 *Strap slippage* With the strap inserted in the buckle, at a point 3 inches from the buckle, the webbing shall be securely attached to a fixed clamp or other suitable device The clamp shall grip the webbing evenly across its entire width At a point 3 inches on the other side of the buckle, there shall be applied a 130-pound pull The pull shall be applied across the entire width of the webbing There shall be no evidence of slippage of the strap through the buckle

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue* One (1) tourniquet, as specified constitutes one unit

5.1.2 *Packaging quantities* The number of units of issue indicated in table I shall be packaged in each unit intermediate and exterior container as applicable, for the required level of protection specified in the procurement document

TABLE I *Packaging quantities*

Unit package	Intermediate package	Exterior container
1 unit	72 units	144 units

5.1.2.1 *Packing variation permitted* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design acceptable to a common carrier, which will insure safe delivery to destination

5.1.3 *Level A*

5.1.3.1 *Unit package* Each unit shall be compactly folded and packaged in a bag fabricated from virgin polyethylene seamless tubing having an average wall thickness of not less than 0 003 inch, with no wall thickness at any point below 0 00225 inch Excess air shall be expelled or evacuated, and enclosure effected by heat sealing

5.1.3.2 *Intermediate package* Intermediate package shall be a box constructed in accordance with Specification PPP-B-636, class 2, compliance symbol W5c or W5s, style RSC, except that the manufacturer's joint shall be effected by the application of a water-resistant tape, or by gluing integral lap to outer face of end or side panel using water-resisting adhesive, in lieu of metal stitches Closure shall be effected by means of water-resistant gummed or pressure-sensitive tape The tape shall be not less than 2 inches wide Three strips shall be applied to both top and bottom so that all seams are covered their full length Tape applied over the lengthwise seams shall extend at least 2 inches onto each of the end panels Tape applied over the end-edge seams shall extend a minimum of 2 inches along the side and top and side and bottom panels

5.1.4 *Level B*

5.1.4.1 *Unit packages* Unit package shall be as specified in 5 1 3 1

5 1.4.2 *Intermediate package* Intermediate package shall be a carton of appropriate size constructed in accordance with Specification PPP-B-636 Closure shall be adequate to prevent spillage of contents under normal handling

5 1.5 *Level C* Units shall be packaged in standard commercial containers of the size and kind commonly used which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America

5.2 Packing

5.2.1 *Level A*

5.2.1.1 *Exterior container* Exterior container shall be a box of appropriate size constructed in accordance with Specification PPP-B-636, class 2 compliance symbol V3c or V3s, style RSC Closure shall be as specified in 30 3 2 1 thereof, and in addition, the top and bottom center seams shall be closed by means of tape conforming to the requirements of Specification PPP-T-45 or PPP-T-76 The tape shall be not less than 3 inches wide and shall extend at least 2 inches onto each of the end panels Strapping will not be required

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5.2.2 *Level B*

5.2.2.1 *Exterior container* Exterior container shall be a box of appropriate size constructed in accordance with Specification PPP-B-636, type I, class 1, and shall be fabricated of not less than 275-pound test fiberboard

5.2.3 *Level C* The subject commodity shall be packed in substantial commercial containers of the type size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order

5.3 Marking.

5.3.1 *Unit package* Each bag shall be marked with stock number, item identification, unit, quantity and name of contractor The required information shall be printed either directly on the bag or on a label inserted within the bag in such manner as to permit ready identification

5.3.2 *Intermediate package* Each intermediate package shall be marked in accordance with Standard MIL-STD-129

5.3.3 *Exterior container* Exterior container shall be marked in accordance with Standard MIL-STD-129

6. NOTES

6.1 The buckle referenced in 3.1.3 shall be North and Judd Mfg Co's No 5105-1½ or equal

6.2 **Ordering data.** Procurement documents should specify the following

(a) Number and date of this specification

(b) When sampling or testing of Government supplied material is required (see 4.1.3)

(3) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

6.3 This specification does not include all types classes, grades, sizes, etc, of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc, which are normally procured to meet Military requirements

6.4 **Transportation description.** Transportation description applicable to this item is

Surgical bandages or dressings in boxes

Carload minimum weight 24 000 pounds subject to Rule 34 of the Uniform Freight Classification

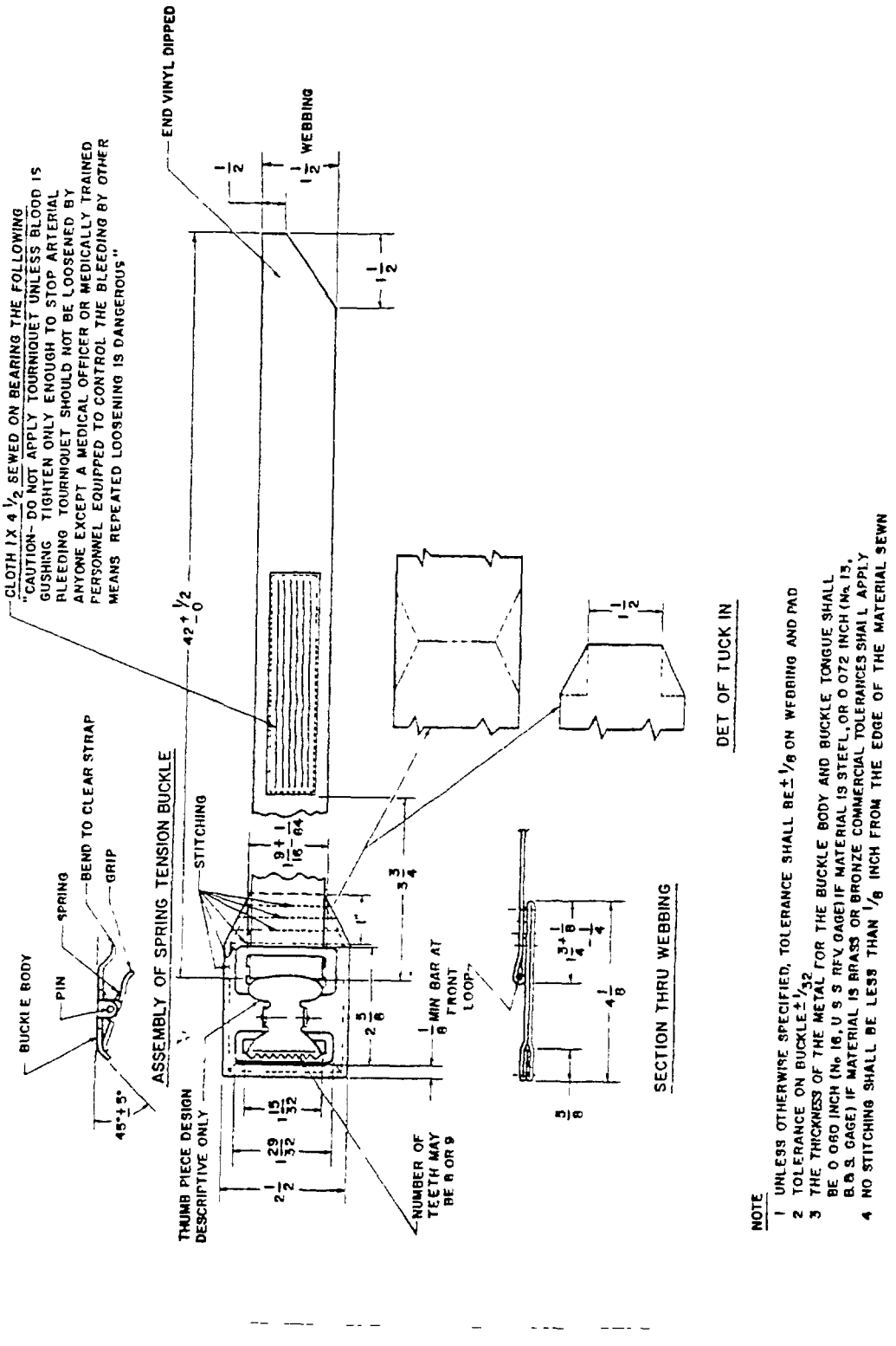
Truckload minimum weight 24,000 pounds per National Motor Freight Classification

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Custodians
Army—MED
Navy—BuMed
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Preparing activity.
Navy—BuMed
(MMSA)

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NOTE

1 UNLESS OTHERWISE SPECIFIED, TOLERANCE SHALL BE ± 1/8 ON WEBBING AND PAD
2 TOLERANCE ON BUCKLE ± 1/32
3 THE THICKNESS OF THE METAL FOR THE BUCKLE BODY AND BUCKLE TONGUE SHALL BE 0 080 INCH (No 16, U S S REV GAGE) IF MATERIAL IS STEEL, OR 0 072 INCH (No 13, B & S GAGE) IF MATERIAL IS BRASS OR BRONZE COMMERCIAL TOLERANCES SHALL APPLY
4 NO STITCHING SHALL BE LESS THAN 1/8 INCH FROM THE EDGE OF THE MATERIAL SEWN

FIGURE I

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